HLTP/x25-32 Microarchitecture Specifications

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0: Preamble

Disclaimer: This project and all related content are **not** intended for any real world application. It is primarily intended as my (the author's) way of learning CPU design, aimed at providing educational use of computer architecture theory. Although trying to imitate real world implementations, some areas may significantly deviate from standard use/understanding; This usually comes down to either intentional design decisions or incompetence (by the author of this document) to fully comprehend architecture design.

Before you read this document, please be aware that it is assumed, that the reader has already familiarized with some basic computer knowledge and additionally ground level knowledge of low level CPU architecture design.

[Placeholder: How this project came to life]

0.1: Technical Terminology, Abbreviations

CPU	Central processing unit
CPI	Clocks per instruction (Multiplicative inverse of Instructions per clock / IPC)
ISA	Instruction set architecture
MSB	Most significant byte
LSB	Least significant byte

Tab. 1: Commonly used abbreviations throughout this document

GPR / GP[X]	General-purpose register [X]
PC	Program counter (commonly also referred to as Instruction Pointer / IP)
IR / IRA / IRB	Instruction register / Instruction register A / Instruction register B
MAR	Memory address register
MDR	Memory data register
SR	Status register
CU	Control unit
ALU	Arithmetic logic unit
MMU	Memory management unit
RAM	Random access memory (sometimes just called “Memory”)
CON / TMEM	Console memory
VMEM	Video memory / Graphics memory

Tab. 1.1: Abbreviations for CPU hardware internals

1: Abstract Design

[Summary of what this chapter covers]

1.1: Common CPU specifications

Word Size	32 bits / 4 bytes
Instruction Width	64 bits / 8 bytes (double word alignment)
Endianness	Big-endian (MSB first, LSB last)
Registers	8x GPRs (1 word), PC (1 word), IRA (1 word), IRB (1 word), SR (1 word), MAR (1 word), MDR (1 word)
Clocks per instruction (CPI)	4
Architecture model	RISC
Pipelining	None
Branch prediction	None (requires Pipelining)
Parallelism	None
Cache architecture	No cache
Virtual memory support	Hardware address mapping, No dynamic remapping during execution
Execution modes	Real / direct mode only, No mode-switching instructions
Extensibility	Not implemented (No co-processing)

Tab. 2: Specifications common among different CPU architectures

2: Hardware

[Brief summary of what this chapter is about]

2.1: Limitations and Design Decisions regarding Emulation

[TODO: Focus more on why the CPU was designed this way because of limitations by emulation; Not what limitations come from the emulation process.]

Since at the current state, the CPU is neither implemented as true hardware, nor concrete enough to be programmable for hardware emulation devices (like FPGAs), there are several limiting factors considering the CPU itself, its environment, and the emulator host system.
First and foremost, the clock speed is currently limited to about 70 – 80 kHz depending on the scenario and host processing power. While this might seem fast enough for most cases, it might be sometimes required for higher speeds to be available. This is especially true for the long term, in which heavy use of screen buffers and live video updates might become necessary, while retaining acceptable latency.
Second, (and this is the primary reason the CPU architecture is limited to 32 bits) is that the CPU is able to handle up to 4 GiB of emulated RAM. This becomes a problem for the host, because without the necessary implementation considerations and parameters, most systems trying to emulate these spaces won't be able to allocate this amount of host memory. This often reduces the effective size of emulated RAM, sometimes causing illegal address errors when trying to develop code for the CPU.

2.2: General Processor Layout

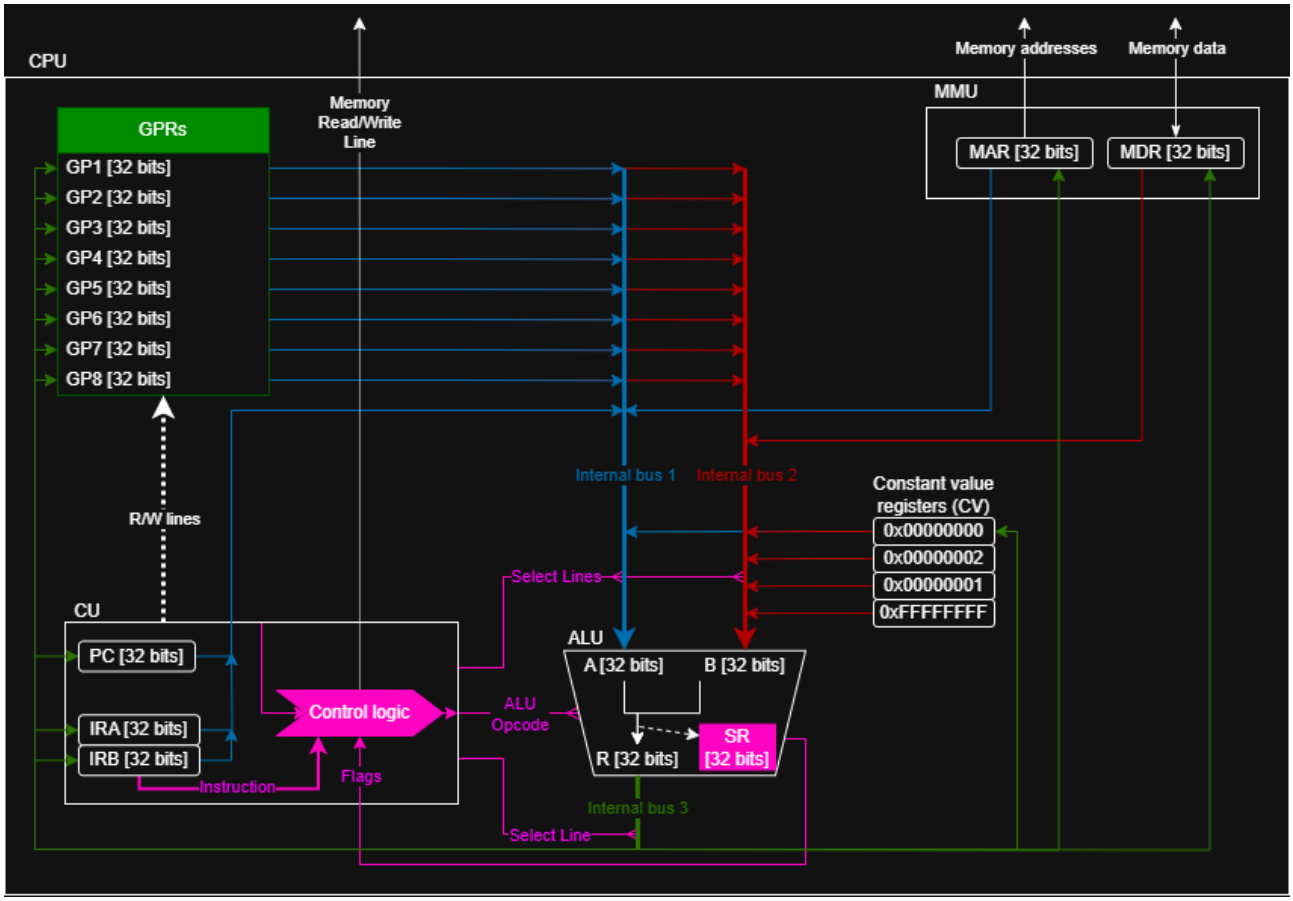

Fig. 1: The x25-32 CPU Hardware Layout as of Processor Revision 0.1-indev

Legend:
Component colors:
Pink: Elements directly affecting, processing or forwarding control signals
Blue: Direct attachments to Internal bus 1
Red: Direct attachments to Internal bus 2
Green: Direct attachments to Internal bus 3
Lines:
Dotted: Single bit line / single “wire”
Solid: Bus (32 bits)
Solid and fat: Bus (32 bits) that changes source / destination depending on control signals (using multiplexers); *Exception: “Instruction” bus from IRA + IRB to Control logic. This line is only fat to emphasize that it carries 2 words from IRA and IRB at the same time, meaning it is 64 bits wide.*
Miscellaneous:
Rounded boxes: Special purpose registers